

Reva

REVA, the Indian electric car has more units produced than any other electric car. It has a top speed of 80kmph, and a range of 80kms on a full 8hr charge

Night vision

Many high-end cars feature night-vision for better and safer driving, enabling the driver to see a longer distance not only at night but also in foggy conditions

As long as there is combustion there will be some unwanted products but it is also not feasible to expect each and every car to turn on a dime and switch to electric or fuel-cell based systems. Alternative fuels require an easier transition, and that is their charm, to significantly reduce emissions while maintaining backwards compatibility. Many current cars can be modified to make them compatible with alternative fuels such as CNG, ethanol, and bio-fuels, and can in fact even be made to run in a dual-fuel mode, with the option to run it off either CNG or gasoline. Furthermore, engines can be specially designed to run on multiple fuels, automatically adjusting their characteristics to those appropriate for the current fuel mixture.

The very word alternative means there's more than one, and there is more than one

Compressed air engines have significant issues that need to be addressed before they enter the market. Such cars are required to be very light, and many fear that the safety of the car may be compromised as lighter, thinner, material will be required for the body. Refuelling and range are also issues. Don't expect to be able to fill up the tank with your trusty old foot pump; they require that the tank be filled up to a pressure of 4500 psi, or over 300 times the atmospheric pressure! Compressed air itself costs energy to make, thus the compressed air engine is only as clean as the energy used to compress the air.

Solar cars

These have oscillated between popularity and obscurity many times. To this day the fact remains that solar energy is just not feasible as a standalone source of energy for running



Hybrid cars such as this *Honda Insight* offer the best balance between fuel and electric

gasoline replacement out there. Although we seem to have settled with CNG today, that is not the end of it, and over time, better alternatives will come up. Hydrogen-powered internal combustion engines are quite possible, and since their end-product is water, the environmental impact is almost nil. Such cars are actively in development.

Air cars

What could possibly be more renewable and less polluting than air itself! Engines which can run on compressed air have been in the making since as far back as 1920, but recently this field has seen significant progress. In-fact TATA is even considering an AirEngine version of its upcoming Nano, and has tied with the Moteur Development International (MDI) in this effort.

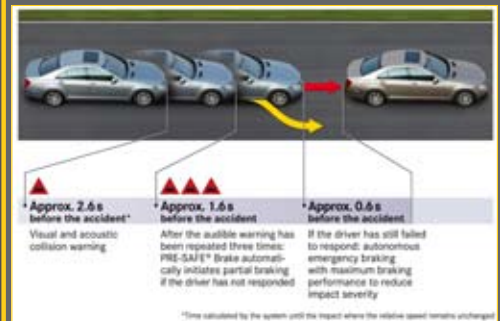
a car. Even the best commercial solutions use solar power as an additional source of energy. Solar energy by its very definition fails to be a fully competent energy source for cars.

A solar car shares the basic concept of an electric car, with additional constraints on the weight of the car body, and other components. To be practical they need a battery as well, to provide a constant power supply to the car even in conditions of low-light. The usual advantages of the electric car such as regenerative braking apply here as well. Some of the available solar technologies for cars are limited to improving the efficiency of electric, hybrid, or even normal cars, by recharging batteries while the car is parked under the sun.

The most popular use of fully solar cars today is in solar-car races, with cars

In cars we trust

The fractions of a second before a crash: Mercedes assistance systems automatically warn, support and apply the brakes if the danger of an accident is acute



*Time calculated by the system until the impact where the relative speed remains unchanged

PRE-SAFE®: Activation of the front belt tensioners in response to radar signals



Unavoidable collision at a junction
Activation of the PRE-SAFE® belt tensioners based on information from the close-range radar



Unavoidable collision with oncoming traffic
Activation of the PRE-SAFE® belt tensioners based on information from the close-range radar

Mercedes-Benz S-Class: BAS PLUS and the DISTRONIC PLUS proximity control assists the driver during emergency brake application. If the driver is distracted and therefore does not register the imminent danger of a rear-end collision, the PRE-SAFE brakes can intervene and automatically brake the car. They act like an "electronic crumple zone".

Mercedes-Benz S-Class: The anticipatory occupant protection system PRE-SAFE® is based in part on the results of real-life accident research. It recognises potentially dangerous situations and activates protective measures for the car's occupants. PRE-SAFE thus builds a link between active and passive safety.